

Lab06. Building infrastructure.

Cloud Programming (W04IST–SI4527G)

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1 Learning goals

1. Building configuration using Terraform.

2 Exercises

Prepare Terraform configuration file to build the following configuration.

Do not use VPC module! Configure each resource individually.

1. Create VPC and subnets.
 1. Create a new VPC with CIDR scope (e.g. 10.0.0.0/16).
 2. Create at least two subnets:
 - public subnet - for resources accessible from Internet,
 - private subnet - for resources without internet access.
 3. Create route tables:
 - for public subnet,
 - for private subnet.
2. Create Internet Gateway and NAT Gateway.
 1. Associate Internet Gateway with VPC.
 2. Add default route to public route table, directing internet traffic to Internet Gateway.
 3. Associate NAT Gateway with public subnet.
 4. Add default route to private route table, directing internet traffic to NAT Gateway.
3. Create two EC2 instances:
 1. Create one EC2 instance in public subnet, with public IP.
 2. Create Security Group and assign it to the first EC2 instance. Add a rule for SSH access from any address.
 3. Create one EC2 instance in private subnet.
 4. Create Security Group and assign it to the second EC2 instance. Add a rule for SSH access from Security Group assigned to the first instance.

2.1 Verification of the configuration

1. Connect to the first EC2 instance using SSH. Verify, if the instance have Internet access. Use commands: `curl`, `ping`, `wget`.
2. Copy your private key to the first EC2 instance. Connect using SSH from the first to the second instance. Verify Internet connectivity.

2.2 Final report

1. Write a report about your configuration in the README.md file.
 1. Explain applied configuration of services, Security groups, Routing tables.
 2. Include most important screenshots presenting your configuration and verification of the connectivity for both instances.

Bibliography

- [1] Y. Brikman, *Terraform: Up and running*, 3rd ed. <https://learning.oreilly.com/library/view/w/terraform-up-and/9781098116736/>; O'Reilly Media, 2022.